

21. (2.75 points)

a. (2 points)

Describe a possible explanation why markets may soften for each of the following theories for the underwriting cycle:

- i. Institutional factors
- ii. Competition
- iii. Capacity and capital levels
- iv. Economic linkages

b. (0.75 point)

Identify two components of an econometric model of the underwriting cycle and briefly explain how the two components are linked.

END OF EXAMINATION

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

QUESTION 21	
TOTAL POINT VALUE: 2.75	LEARNING OBJECTIVE(S): C8
SAMPLE ANSWERS	
Part a: 2 points	
<i>Sample responses are separated into the four theories listed in the question:</i>	
Theory i • There is a data lag as pricing indications are done on historical data which might indicate a rate decrease, however the rate may be inadequate but due to the lag we don't see it in the data yet. A further rate decrease brings rates too low. • Data lags can make insurers slow to react to changes in trends. Using lagged data with low frequency and severity trends could cause prices to fall even when more recent trends (not contemplated yet) have risen. Therefore, business is being written below adequacy. • There is a lag in data aggregation by the statistical bureau, creating a lag in rate regulation, which may result in the regulator approving inadequate rates. • Regulators may refuse to approve or take a long time to approve necessary rate increases, resulting in insurers having to keep prices low. Theory ii • In competitive markets, companies may undercut each other on price in order to gain market share. This can happen especially when an insurance product is new and there is little regulation or data. • A new entrant may enter without much experience. They may be priced lower than other more experienced or sophisticated companies. The new entrant will increase the supply and drive down the price for insurance. Theory iii • With more capital through a capital infusion or otherwise, insurers have more surplus and can afford to write more business. When supply increases, the price decreases. • When insurers have more capital, they have increased capacity and need to write more premium to earn their target ROEs. Theory iv • An increase in interest rates in the market will increase the investment income insurers earn so they will increase their supply because they are able to make enough money even with lower underwriting income. • If GDP shrinks, unemployment rises, or there is a general economic recession, demand drops because insureds have fewer assets to protect and diminished purchasing power. If supply stays about the same, this would result in a lower equilibrium price. • A large, unexpected increase in inflation will increase claim costs, resulting in inadequate rates.	

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Part b: 0.75 point

- Capital and supply of insurance. Capital infusion will give companies more capacity to write more policies, increasing supply. Shocks that reduce capital (e.g. cat event) will reduce capacity and decrease supply.
- Capital and demand curve. If customers are sensitive to quality, demand will increase after a capital infusion since the insurers have more capital and are less likely to go insolvent.
- Profit flows and capital. When the industry is profitable, capital increases due to retained earnings and capital infusions. When the industry is not profitable, capital decreases due to operating losses and exits.
- Supply and demand curves. They are linked because their intersection determines the equilibrium price and quantity of insurance.
- Supply and demand curves. They are linked by capital supply which can increase with capital infusions or decrease with shocks (such as catastrophe events). When capital rises, supply curve shifts to the right since insurers have increase capacity to write business. Demand also shifts to the right (to a smaller extent) due to perception of increase insurance quality. Overall, price decreases. The opposite movements occur when a shock happens.
- Statistical validity of the technical approach and recognition of human factors from the soft approach. The two components are linked by analyzing supply and demand, capital capacity, etc. with the incorporation of expert opinions to model the underwriting cycle.

EXAMINER'S REPORT

Candidates were expected to know four theories of the underwriting cycle and describe how each theory would explain why markets soften. Candidates were expected to provide an example of how two components function in an econometric model of the underwriting cycle.

Part a

Theory i

Candidates were expected to know that the uncertainty inherent in projecting future losses, which may be exacerbated by data reporting and regulatory delays, can result in inadequate pricing.

Common mistakes included:

- Responses related to management action/underwriter incentive plans, or changes in technology.

Theory ii

Candidates were expected to know that companies may charge lower prices due to an optimistic view of the future or a desire to increase market share, and that competitive markets tend to move to the lowest price offered (the "winner's curse").

Theory iii

Candidates were expected to know that an increase in industry capital or capacity would lead to an increase in the supply of insurance, resulting in a lower equilibrium price.

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Theory iv

Candidates were expected to explain how certain general economic factors can lead to a softening of insurance prices.

Common mistakes included:

- Vague responses that failed to sufficiently describe how a specific factor would close lower market prices.

Common mistakes included:

- Describing what would cause market hardening rather than softening, or market fluctuations rather than softening specifically.
- Describing what would cause lower premiums but not lower premium adequacy.

Part b

Candidates were expected to either:

- Name two variables used in the econometric model and describe how they are related to or influence each other; or
- Describe how the econometric model blends the soft and technical styles of modeling.

Common mistakes included:

- Inaccurately depicting how an econometric model functions.
- Naming a specific technical modeling method and soft modeling method as the two components.
- Describing a time series model.